Simulated Drives over Measured Road Profiles for Car-Based IRI Estimation: What Transfers and What Needs Calibration

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***Abstract*—Estimating the International Roughness Index (IRI) from the vibration of ordinary cars needs paired data: drives of the car type over road sections with surveyed profiles. Vehicle simulation can generate such pairs from the profiles alone. We test what estimators trained on simulated drives transfer to a real car, using public profilometer-referenced data from one car model (LiRA-CD, four routes, 168 km), routes held out from training and 5 km-block bootstrap intervals. Gradient boosting trained on real drives reaches a within-route mean absolute error (MAE) of 0.39 m/km. Trained only on a randomised half-car simulation over the measured profiles of the other routes, it ranks segments within routes as well under each of nine sensor and vehicle models (Spearman 0.82–0.84 vs 0.83), but so does an untrained, speed-normalised vibration RMS (0.82). What has to be learned is the level, and the zero-shot bias ranges from -0.09 to $+0.43$ m/km: the pre-specified simulator matched it (MAE 0.42 m/km) only because half of its passes carried a phone-holder resonance, and across variants the bias follows a deficit of simulated vibration at 10–15 Hz. A level offset from 5 km of paired drives brought every variant to 0.42–0.56 m/km, better than training on those 5 km alone (0.68 m/km) in 13–20 of 20 draws; with 10 km or more, real drives alone beat the three variants with the largest deficit. Simulated vibration also grows faster with speed than the real car's. With a short calibration, simulation can stand in for most paired drives, but how many it saves depends on how well the simulated vibration matches the car, which we could test for one car only.**

***Index Terms*—International Roughness Index, road roughness, vehicle vibration, connected vehicles, simulation-to-real transfer, calibration.**

I. Introduction

The International Roughness Index (IRI) is the standard summary of longitudinal road roughness [1], [2]. Measuring it requires a longitudinal profile from a profilometer, so minor roads are surveyed rarely. Production cars that already carry accelerometers could fill the gap [3], [4], [5], but their vibration depends on the vehicle, the sensor mounting and the speed, so the mapping from vibration to IRI has to be established for each car type from paired data: drives of that car over road sections whose profile was measured [6]. Collecting such paired drives is the expensive step.

Vehicle simulation offers another source of pairs. Driving a vehicle model over profiles that road agencies already survey yields vibration with exact IRI labels, at any speed and for many vehicle parameter sets. Estimators trained this way have been validated in simulation [7], [8] or, for roughness classes, with a real-vehicle test [9], and calibrated physical vehicle models can invert real car vibration into road profiles [10], [11]. What is missing is evidence on real cars, on roads not used for training, of what a simulation-trained IRI estimator actually transfers and how much paired real data it still needs.

We study this question on LiRA-CD [12], public data of one car model on four routes with profilometer references. Estimators trained on real cars are compared with the same estimators trained only on simulated drives over the measured profiles of the other routes. We vary the sensor and vehicle model of the simulator, calibrate its output with k km of real data, compare simulated and real drives at equal profile budget, measure speed dependence on real roads with repeated passes over the same segments, and compare simulated and real vibration spectra on the same road segments at the same speeds.

The answer has three parts. Ranking segments within a route does not discriminate: simulation-trained gradient boosting ranks them as well as the real-trained model under every simulator variant, but so does an untrained, speed-normalised vibration RMS. What has to be learned is the level, and it does not transfer reliably: it depends on the simulated sensor path and follows a deficit of simulated vibration at 10–15 Hz, and the pre-specified simulator matched it only through a sensor component (a phone-holder resonance) that was not designed to model the car's sensor. A few kilometres of paired drives fix the level: with a level offset from 5 km of paired drives, simulation over all profiled roads beats training on those 5 km for every variant, and for six of the nine variants it comes close to real training on all routes; with 10 km or more, real drives alone beat the variants whose vibration matches the car worst. Speed dependence transfers only in part, and the simulated vibration scales with speed differently from the real car's.

Our contributions are: (i) a route-disjoint benchmark on public profilometer-referenced car data with dependence-aware inference and baselines from response-type physics (a calibrated quarter-car inversion), learned estimators and

simulation-trained estimators (Section V-A); (ii) a decomposition of what simulation transfers to a real car across nine sensor and vehicle models (for most of them the vibration-to-IRI mapping up to a level offset, not the level), with a sensor factorial and a spectral comparison on the same road segments that locate the level error (Sections V-B and V-C); (iii) the amount of paired real data that simulation replaces, at equal profile budget and with level calibration, for every simulator variant (Section V-D); (iv) fixed-road speed elasticities of real- and simulation-trained estimators measured on real roads (Section V-E). The study revises an earlier, unpublished pipeline of ours trained on a game engine; its label and protocol problems are documented in Supplement S2, and its corpus serves here only as a negative reference. Code, protocols, configurations and all predictions are provided with the paper.

II. Related Work

Response-type measurement and physical vehicle models. IRI is defined on the longitudinal profile through a reference quarter car (the "Golden Car") driven at 80 km/h [1], [13], [2]. Response-type systems measure a vehicle's reaction instead of the profile and must be correlated with a reference profiler for each vehicle [6]. Model-based methods estimate roughness from the responses of a specific vehicle with fixed or identified vehicle models and transfer functions [14], [15]; Bridgelall's road impact factor normalises vertical-acceleration energy for connected vehicles [16]. Closest to our data, Skar and Levenberg [10] invert road profiles from the accelerations of ten nominally identical electric cars with a quarter-car model, a PID controller and nonlinear error minimisation, and compare them with a laser profilometer; Agebjär et al. [11] identify a vehicle model of a production car and reconstruct the profile with a Kalman filter. These methods need a model of each vehicle with known or identified parameters. Our baseline M7 follows the same idea with a quarter car calibrated on the training routes; the vehicle parameters calibrated in [10] are not available to us (its open code contains only the Golden-Car parameters), so M7 is not a reproduction.

Learned estimators on real data. Smartphone studies report agreement with reference roughness that depends on speed, vehicle and phone [17], on mounting [18] and on the number of repeated runs [19]; see also [20], [3]. Data-driven models accounting for speed, vehicle type and mounting were trained on real smartphone data [4], and semi-supervised learning exploits unlabelled responses of crowdsourced vehicles [21], [22]. On LiRA data, Bajic et al. [23] trained regressors on the car's vertical acceleration and speed along the M3 motorway and tested them on the last 20 % of the same road; an evaluation framework for smartphone systems is proposed in [24]. We test on routes not used for training and report uncertainty that accounts for spatial dependence.

Learning from simulated responses. Neural networks trained on simulated vehicle responses have been used to reconstruct profiles and classify roughness [25], to classify roughness grades from quarter-car simulations [8] and, with simulation and a real-vehicle test, from the unsprung-mass response [9], and, closest to our question, to estimate IRI without vehicle-specific calibration: Jeong et al. [7] trained CNNs on simulated responses of four vehicle types with real driving-speed profiles and validated them numerically with real road IRI information. Domain randomisation is the standard device for transferring models from simulation to real data [26]. Game engines provide vehicle dynamics for software testing [27], [28], but IRI labels derived from their authored road meshes need validation (Supplement S2). We add a real-car, route-disjoint test of simulation-trained estimators, a decomposition of what transfers (ranking, level, speed dependence), and the amount of paired real data still needed.

III. Data and Simulation

Real cars: LiRA-CD. LiRA-CD [12], [29] contains Green-Mobility Renault Zoe cars instrumented with an AutoPi telematics unit (three-axis accelerometer, GPS) connected to the CAN bus, and surveys of the same lanes by the Danish Road Directorate's P79 profilometer. We use the four routes with car data: two urban roads in Copenhagen (CPH1, part of the O2 ring road, and CPH6) and two motorways (M3, M13), each in both directions. The P79 files provide the wheel-path laser elevations at 0.1 m with GPS. In the car files the accelerometer is logged at a nominal 52.6 Hz (66 % of passes) or 333 Hz, quantised to 0.01 g in most passes, with a median mounting tilt of 10°; speed comes from the CAN bus and position from GPS at 0.9–3.4 Hz. After map matching (Section IV) the data contain 16,293 segment-passes over 1,678 distinct 100 m segments (168 km) from 91 passes of at least 8 cars (telematics units). Urban segments have mean IRI 3.36 m/km and are driven at 43 km/h on average, motorway segments 1.16 m/km at 87 km/h. The same cars drive several routes, so leave-one-route-out evaluation is route-disjoint but not vehicle-disjoint for estimators trained on real data (Section V-A tests the consequence).

Labels. Labels are the IRI per 100 m, mean of both wheel paths, recomputed from the P79 profiles with an implementation of the quarter-car algorithm [13], [2] that reproduces the P79's own IRI ($r = 1.000$ over 1,520 segments; an algorithm check, since both come from the same profiles). Labels derived from authored game-engine geometry are not equivalent: the curvature-based mesh correction of our earlier pipeline lowers the IRI of these real urban profiles by 28 % (Supplement S2).

Parametric vehicle simulator. A linear four-degree-of-freedom half-car (bounce, pitch, front and rear unsprung masses) is driven by a road profile under the front axle and the same profile delayed by the wheelbase under the rear axle, after a 250 mm tyre-envelope moving average; on measured profiles the input is the mean of both wheel paths (no roll), and simulated units are labelled exactly like the real ones. The solver deviates by at most $4.4 \cdot 10^{-9}$ (relative) from a fine-step Runge–Kutta integration of the continuous system. Vehicle parameters are drawn from ranges for passenger cars [30] (sprung mass 1000–1800 kg, wheelbase 2.4–3.0 m, body frequency 1.0–1.6 Hz, damping ratio 0.20–0.45; Table S3), speed

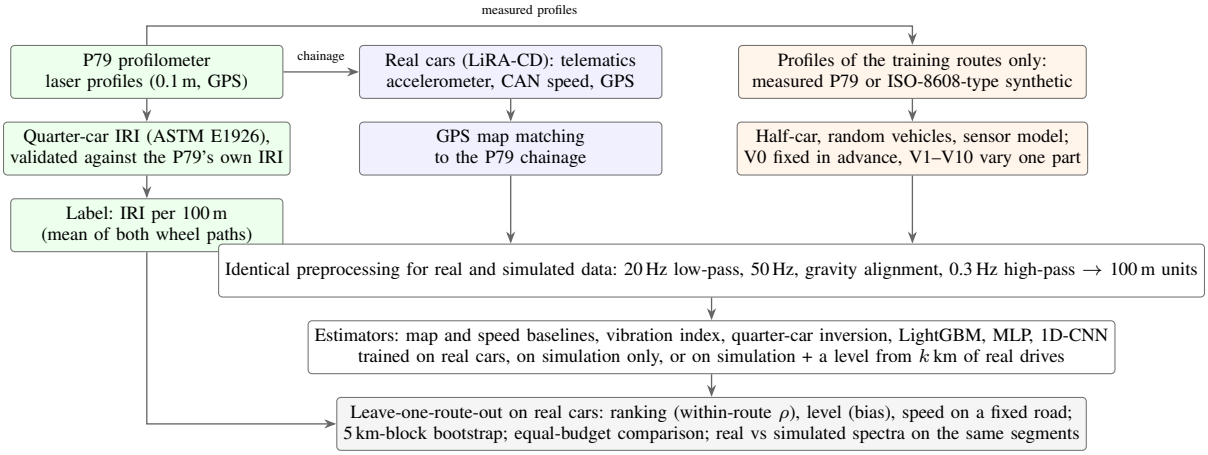


Fig. 1. Study design. Labels are recomputed from the P79 profilometer's laser profiles with an IRI implementation validated against the profilometer's own output; simulated units are labelled with the same code. Real and simulated vibration go through identical preprocessing, and every estimator is evaluated on real routes it never saw; simulated drives never use the profile of the test route.

independently of roughness (20–110 km/h with fluctuations and braking events). The sensor model applies an optional holder resonance, an anti-aliasing filter, a mounting rotation, irregular sampling, scale and bias errors, white noise, quantisation and sample loss. Simulated and real signals then go through identical preprocessing code.

Simulator variants. The reference simulator **V0** was fixed before any real test result: 30 random vehicles, measured P79 profiles of the training routes, and a 50/50 mixture of a telematics-like sensor (55–62 Hz, 0.01 g, rigid mount) and a phone-like sensor (90–210 Hz, 0.6 mg) with a holder resonance at 15–40 Hz. To find out which modelling choices matter, we compare it with ten variants, listed in a protocol written after an independent review and before the analyses reported here were run (the zero-shot results of V3–V6 were known from our earlier analysis and those of V1 and V2 from the review, so this comparison is diagnostic, not confirmatory): the telematics-like half alone (**V1**) and the phone-like half alone (**V2**) of V0; the two missing cells of a 2×2 design, telematics-like with the holder resonance (**V7**) and phone-like without it (**V8**); an ideal sensor (**V3**); a single telematics sensor matched to statistics measured on the unlabelled LiRA-CD streams (**V5**) and its 50/50 mixture with the phone-like sensor (**V6**); one mid-range vehicle instead of 30 (**V4**); and synthetic ISO-8608-type profiles [31] alone (**V9**) or added to the measured ones (**V10**). The simulated drives never use the profile of the test route.

Earlier BeamNG corpus. As a negative reference we also train on the corpus of our earlier study: 19 manually driven trips of 3 BeamNG.tech vehicles on 3 maps, with labels from ray-cast elevation; its acquisition duplicated 48–68 % of IMU samples (Supplement S2).

IV. Estimators and Evaluation

Preprocessing and features. Real and simulated accelerometer streams go through the same code: interpolation to 200 Hz, zero-phase 20 Hz low-pass, decimation to 50 Hz, projection on the pass-mean acceleration direction (removing the mounting tilt), and a 0.3 Hz high-pass. LiRA-CD GPS fixes are matched to the P79 track of the same route and direction with a heading constraint. A unit is a 100 m segment traversed once by one pass (≥90 % coverage). Each unit is described by speed (log mean, coefficient of variation) and vibration features of the dynamic vertical acceleration (log RMS, 95th percentile and maximum of the magnitude, crest factor, kurtosis, band RMS in 0.5–2, 2–5, 5–10 and 10–15 Hz, RMS above 1 Hz, RMS at wavelengths below 5 m).

Estimators. All learned estimators predict log IRI. *Without vibration*: M0, the training median; M0b, a road-class prior that predicts each held-out route with the median IRI of the other route of the same class (needs only a map); M1, LightGBM [32] on the two speed features. *Physics-based*: M2c, a vibration index $\log \mathrm{IRI} = a + b \log(\mathrm{RMS}/v^{\alpha})$ with α fitted; M7, a calibrated quarter-car inversion: the empirical transmissibility of the car and sensor (acceleration spectrum divided by the road displacement spectrum at the pass speed) is estimated on the training routes, a quarter car with sensor gain is fitted to it, each test unit's road spectrum is obtained by spectral division, and IRI follows from the Golden-Car response to that spectrum under a Gaussian assumption (the formula reproduces the validated IRI from true profile spectra with $r = 0.94$ in log scale); M7 is also reported with a log-linear output calibration fitted on the training routes. *Learned*: M3, LightGBM on all features (also without speed inputs); M4, an MLP; M5, a 1D-CNN on the resampled acceleration (the architecture of our earlier study); neural nets are averaged over five seeds. *Simulation-trained*: M3 and M2c fitted only to simulated units of a simulator variant (zero-shot). LightGBM uses one of six fixed configurations chosen by inner leave-one-route-out validation when real training routes are available, and its first configuration otherwise.

Level calibration and value of simulation. A simulation-trained estimator is calibrated with k km of real data from the

training routes (contiguous 1 km blocks, 20 random draws per k): either a level offset only (the median log ratio of label to prediction over the calibration passes) or an affine map in log space. To measure what simulation replaces, real-only and simulation-only training are compared at *equal profile budget*: for the same draw of k km, LightGBM is trained either on the real passes over those blocks (which need the profiles and paired car drives) or on simulated passes over the profiles of the same blocks (which need the profiles only).

Speed on a fixed road. Each segment is driven about ten times at different speeds, so the speed dependence of an estimator can be measured on real roads without confounding by roughness: the within-segment elasticity is the slope of the pass-level log prediction on log speed with segment fixed effects. A roughness estimator should have elasticity 0; our earlier criterion of $\pm 10\,\%$ between 40 and 100 km/h corresponds to $|\text{elasticity}| \leq 0.104$. For the raw vibration RMS the elasticity is the car's physical speed exponent.

Simulated vs real vibration. Welch spectra of the preprocessed acceleration (2.56 s windows) are compared between real passes and re-simulated passes of four sensor variants and the ideal sensor over the same P79 segments, on cells of segment $\times$ 10 km/h speed band present in both. For all nine measured-profile variants we compare the band RMS features on the same cells (real/simulated amplitude ratio) and the within-segment speed elasticity of the raw RMS, using only simulated passes within the speed range of the real passes over each segment.

Evaluation and inference. The outer split is leave-one-route-out over {CPH1, CPH6, M3, M13}; both directions of the test route are held out together, and no test-route data enter training, calibration or model selection. The evaluation unit is the 100 m segment, predicted by the mean of its pass predictions. We report MAE (primary), bias and Spearman's ρ within each route and averaged over routes, and pooled values; partial ρ given segment speed measures ranking beyond speed. Segment errors are spatially correlated (residual autocorrelation 0.44 at 100 m, 0.11 at 1 km), so paired differences between estimators are bootstrapped over 5 km blocks (40 clusters, 2000 resamples), and a difference is called supported only if that interval excludes zero; segment- and 1 km-level intervals and a route-level interval (t with 3 degrees of freedom) are in Table S5. With four routes, no claim about route-to-route variation is inferential. All intervals are conditional on the trained models (one per fold); training variability enters only through the seeds of the neural nets and the draws of the calibration analyses.

Evidence classes. The estimators, simulator V0, splits and metrics were fixed before any test result (class C1). After an independent review we wrote a second protocol and only then ran the analyses of the simulator variants, calibration, equal-budget comparison, fixed-road speed analysis, spectra and baselines M0b and M7. These analyses use routes whose results were already known, so we label them as re-analyses (R) or diagnostics (E) and do not use them to select a configuration; the vehicle-number experiment in Section V-F uses freshly simulated test data (S). Four diagnostics were added to that protocol after some of its results were known (the equal-budget comparison for V1–V8, draw-by-draw comparisons with real-only training, the speed scaling of raw vibration and paired elasticity differences); they are reported for every variant (Supplement S1).

V. Results

A. Real cars

Table I lists every estimator on the held-out routes (per-route values in Table S4). A map alone (M0b) or speed alone (M1) gives within-route MAEs of 0.75 and 0.69 m/km; speed ranks segments only weakly within a route ($\rho = 0.39$). The calibrated quarter-car inversion M7 ranks segments as well as the learned estimators ($\rho = 0.83$) but has a larger error (paired difference to LightGBM $+0.081$ [$+0.043$, $+0.115$] m/km), and its output calibration on the training routes made it worse. LightGBM and the MLP differ negligibly ($+0.004$ [-0.013, $+0.023$], although the MLP is worse on all four routes); the 1D-CNN of our earlier study is worse by $+0.045$ [$+0.019$, $+0.075$], almost entirely on CPH1 ($+0.14$ m/km there, between -0.02 and $+0.02$ on the other routes). Errors differ by a factor of four to five between road classes: LightGBM has MAE 0.60 and 0.70 m/km on the urban routes and 0.13 and 0.15 m/km on the motorways.

Two checks concern the benchmark itself. Because the same telematics units drive several routes, we retrained LightGBM only on units that never drove the held-out route; its MAE (CPH1 0.63, CPH6 0.69, M13 0.15 m/km) equals that of size-matched random subsets (0.63, 0.70, 0.15; Table S7), so vehicle overlap did not measurably favour the real-trained estimators (the M3 fold is not interpretable: only urban training units remain). On the within-road split of Bajic et al. [23] (first 80 % of M3 for training, last 20 % for testing), LightGBM reaches MAE 0.115 m/km and $R^2 = 0.81$ (Table S8); the numbers of that and our protocol answer different questions and are not compared further.

B. Zero-shot transfer: the level does not transfer

Trained only on simulated drives over the profiles of the three training routes, the pre-specified simulator V0 reaches a within-route MAE of 0.42 m/km against 0.39 for the real-trained model. The paired difference, $+0.019$ m/km, has a 5 km-block interval of [-0.003, $+0.040$]; at the segment level, which ignores spatial dependence, the interval [$+0.008$, $+0.031$] would have declared the real-trained model better (Table S5). The ranking of segments within routes is the same ($\rho = 0.83$ vs 0.83; partial ρ given speed 0.81 vs 0.81). Within-route ranking does not discriminate between estimators, however: the untrained statistic $\text{RMS}/v^{0.5}$ ranks segments as well ($\rho = 0.82$, partial 0.81; Table S4), since a Spearman correlation within a route is unaffected by any monotone calibration. What training has to supply is the level of the mapping from vibration to IRI.

V0's agreement in level is not a robust property of simulating over measured profiles (Table II, Fig. 2). Across

TABLE I

REAL CARS (LIRA-CD), LEAVE-ONE-ROUTE-OUT, 100 M SEGMENTS. WITHIN-ROUTE METRICS ARE COMPUTED PER HELD-OUT ROUTE AND AVERAGED OVER THE FOUR ROUTES; "PARTIAL ρ": SPEARMAN CORRELATION WITH IRI GIVEN SEGMENT MEAN SPEED. ΔMAE: PAIRED DIFFERENCE TO THE REAL-TRAINED LIGHTGBM WITH A 95 % CLUSTER-BOOTSTRAP INTERVAL OVER 5 KM BLOCKS (40 CLUSTERS); "ROUTES": NUMBER OF HELD-OUT ROUTES ON WHICH THE ESTIMATOR IS BETTER. NEURAL NETS: 5-SEED ENSEMBLES. M7 IS EVALUATED ON THE SEGMENTS WITH A USABLE ROAD SPECTRUM.

Estimator	Within route			Pooled		ΔMAE vs M3	Routes
	MAE	ρ	partial ρ	MAE	bias	[5 km-block CI]	better
No vibration input							
M0 constant (training median)	1.56	–	–	1.67	−0.64	+1.249 [+1.118, +1.387]	0/4
M0b road-class prior (map only)	0.75	–	–	0.81	−0.20	+0.387 [+0.307, +0.475]	0/4
M1 speed only	0.69	0.39	0.19	0.74	−0.33	+0.324 [+0.252, +0.404]	0/4
Physics-based, calibrated on real training routes							
M2c vibration index	0.62	0.76	0.77	0.67	−0.13	+0.251 [+0.202, +0.306]	0/4
M7 quarter-car inversion	0.46	0.83	0.81	0.52	−0.09	+0.081 [+0.043, +0.115]	0/4
M7 + output calibration	0.57	0.83	0.81	0.64	+0.09	+0.197 [+0.134, +0.257]	0/4
Learned on real cars (3 other routes)							
M3 LightGBM (reference)	0.39	0.83	0.81	0.42	−0.11	ref.	–
M3 LightGBM, no speed input	0.62	0.76	0.72	0.65	−0.24	+0.235 [+0.177, +0.294]	0/4
M4 MLP	0.40	0.83	0.81	0.42	−0.09	+0.004 [−0.013, +0.023]	0/4
M5 1D-CNN	0.43	0.81	0.79	0.47	−0.21	+0.045 [+0.019, +0.075]	1/4
Learned on simulation only (zero-shot)							
V0 LightGBM, measured profiles	0.42	0.83	0.81	0.44	+0.01	+0.019 [−0.003, +0.040]	1/4
V10 LightGBM, measured + synthetic	0.44	0.84	0.82	0.46	−0.04	+0.039 [+0.007, +0.068]	2/4

TABLE II

SIMULATOR VARIANTS: LIGHTGBM TRAINED ONLY ON SIMULATED DRIVES (ZERO-SHOT), EVALUATED ON THE HELD-OUT REAL ROUTES. BIAS BY ROAD CLASS; WITHIN-ROUTE ρ; SCALE b: SLOPE OF LOG PREDICTION ON LOG IRI WITH ROUTE FIXED EFFECTS (REAL-TRAINED: 0.66); SPEED ELASTICITY OF THE ESTIMATE ON A FIXED ROAD (SECTION V-E); THEN MAE AFTER ESTIMATING ONLY A LEVEL OFFSET FROM k KM OF REAL DATA OF THE TRAINING ROUTES (MEAN OVER 20 DRAWS; 2.5–97.5 % RANGE IN BRACKETS FOR 5 KM). DIAGNOSTIC VARIANTS (CLASS E) EXCEPT V0, WHICH WAS FIXED BEFORE ANY TEST RESULT.

Simulator variant	Zero-shot	Bias (m/km)		Within-	Scale	Speed	+ level from k km real		
	MAE	urban	motorway	route ρ	b	elasticity	1 km	5 km	20 km
V0 mixture (reference, pre-specified)	0.44	−0.12	+0.17	0.83	0.69	−0.06	0.49	0.42 [0.40, 0.45]	0.42
V1 telematics-like, rigid	0.68	+0.32	+0.56	0.83	0.66	+0.02	0.59	0.53 [0.47, 0.60]	0.54
V7 telematics-like + holder	0.46	−0.05	+0.23	0.83	0.70	−0.15	0.50	0.43 [0.40, 0.46]	0.43
V8 phone-like, rigid	0.55	+0.21	+0.34	0.84	0.71	−0.00	0.51	0.45 [0.42, 0.49]	0.46
V2 phone-like + holder	0.41	−0.22	+0.06	0.83	0.69	−0.10	0.48	0.42 [0.40, 0.45]	0.41
V3 ideal sensor	0.67	+0.27	+0.56	0.83	0.62	+0.01	0.60	0.54 [0.48, 0.62]	0.55
V5 telematics matched to LiRA-CD	0.69	+0.25	+0.60	0.82	0.64	+0.01	0.62	0.56 [0.51, 0.64]	0.58
V6 matched telematics + phone-like	0.46	−0.24	+0.22	0.83	0.69	+0.06	0.52	0.45 [0.41, 0.48]	0.45
V4 one mid-range vehicle	0.63	+0.39	+0.41	0.83	0.66	−0.05	0.50	0.44 [0.41, 0.46]	0.43
V10 measured + synthetic profiles	0.46	−0.26	+0.23	0.84	0.66	−0.03	0.53	0.46 [0.41, 0.50]	0.46
V9 synthetic profiles only	0.71	+0.09	+0.70	0.81	0.66	+0.23	0.84	0.75 [0.65, 0.88]	0.77
trained on real cars (M3)	0.42	−0.19	−0.02	0.83	0.66	−0.15	–	–	–

the nine variants on measured profiles (V0–V8), the within-route ρ stays between 0.82 and 0.84, the partial ρ between 0.80 and 0.82 and the scale b between 0.62 and 0.71 (real-trained: 0.66), whereas the pooled zero-shot bias ranges from -0.09 to $+0.43$ m/km and the MAE from 0.41 to 0.69. The 2×2 sensor design shows what set V0's level: adding the holder resonance changes the bias by -0.35 m/km for the telematics-like and by -0.36 m/km for the phone-like sensor, switching the sensor family by -0.16 (rigid) and -0.17 m/km (with holder). V0 thus owes its level mainly to the holder resonance of its phone-like half, a component meant to model phone holders rather than the car's telematics unit. An ideal sensor (V3) and a telematics sensor matched to the rates, quantisation, noise and tilt measured on the LiRA-CD streams (V5) both overestimate by about 0.4 m/km, and so does a single mid-range vehicle (V4). Synthetic profiles alone (V9) still rank segments ($\rho = 0.81$) but overestimate on motorways (bias $+0.70$ m/km); adding them to the measured profiles (V10) changes little. The earlier game-engine corpus does not transfer under this protocol (within-route $\rho = 0.35$, MAE 1.36 m/km; Supplement S2).

C. Where the simulated vibration differs

On the same segments and speed bands, the real car's vibration differs from all simulated variants in the same way below 10 Hz and differently above (Fig. 3a, Table S15): all variants lack energy at 2–5 Hz (real/simulated power 2.4–3.6) and match it at 5–10 Hz, whereas at 10–15 Hz the real car has 5.2 and 4.9 times the power of the ideal and the rigid telematics-like sensor but only 1.9 times that of the phone-

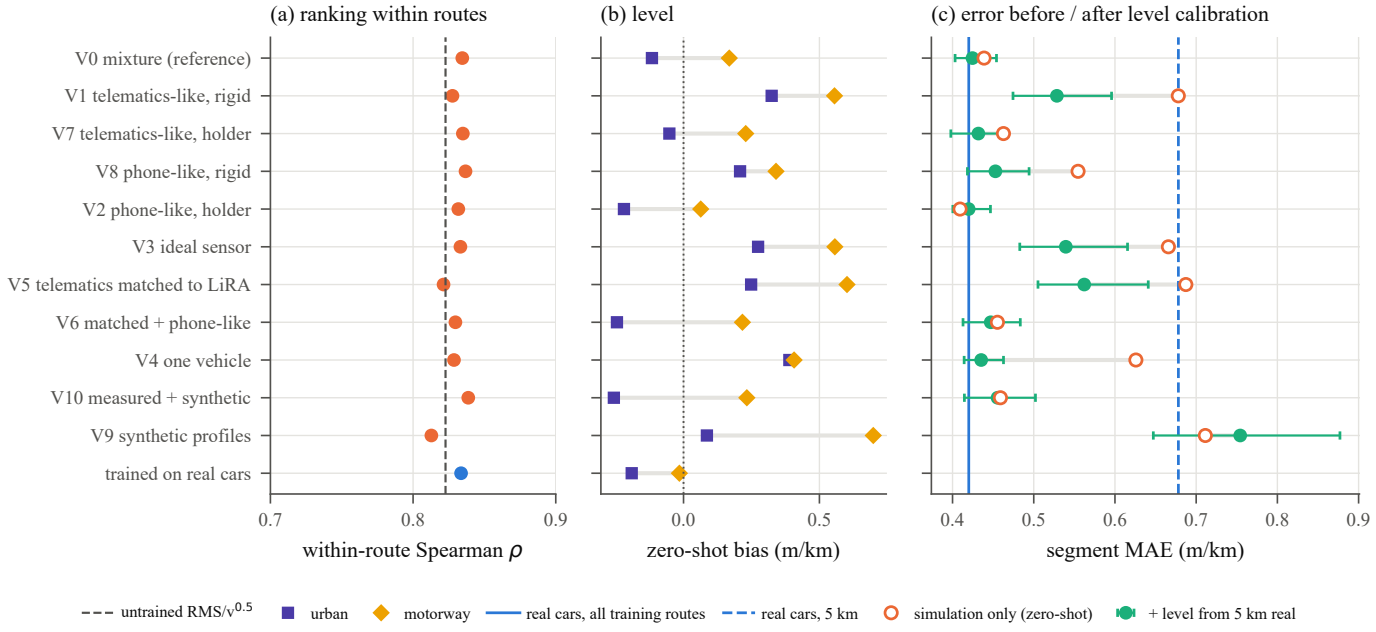


Fig. 2. Simulator variants, LightGBM trained on simulated drives only and tested on the held-out real routes. (a) Within-route Spearman ρ (real-trained model in blue; dashed: the untrained statistic $\mathrm{RMS}/v^{0.5}$). (b) Zero-shot bias on urban and motorway segments. (c) Pooled segment MAE zero-shot (open circles) and after a level offset estimated from 5 km of real passes of the training routes (filled; bars: 2.5–97.5 % range over 20 draws); solid line: trained on all real training data; dashed: trained on 5 km of real data.

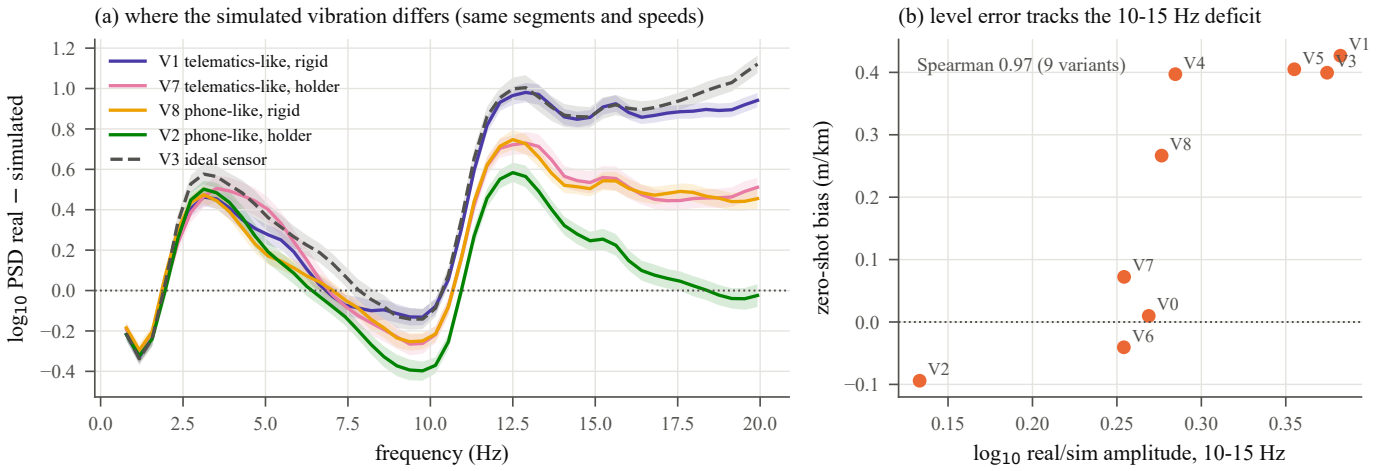


Fig. 3. (a) Mean difference of $\log_{10}$ power spectral density, real minus simulated, on the same segments in the same 10 km/h speed bands (re-simulated passes of four sensor variants and the ideal sensor; bands: 95 % intervals from a 1 km-block bootstrap). (b) Zero-shot bias of the nine measured-profile variants against the real/simulated amplitude ratio of the 10–15 Hz band ($\log_{10}$).

like sensor with holder resonance. Across the nine variants the zero-shot bias orders like this 10–15 Hz deficit (Spearman 0.97; Fig. 3b), as expected if an estimator trained on too little simulated vibration reads the car's extra vibration as roughness, not like the deficits at 0.5–2 or 2–5 Hz (0.22 and 0.25), and the deficit is larger on motorway than on urban segments for 9 of 9 variants (amplitude ratio 1.39–2.81 vs 1.33–1.98; Table S16), as is the bias. The real car's empirical transmissibility peaks at 12.5 Hz (Fig. S3b). Passes logged at 333 Hz contain more, not less, 10–15 Hz power than passes logged at 52.6 Hz on the same cells (−0.24 in $\log_{10}$, confounded with the device), so the excess is not an aliasing artefact of the slower logging; its physical source is not identified (the half-car has no roll or body flexibility, and the mounting of the telematics unit and the unsprung dynamics of the car are unknown).

A second mismatch is common to all variants: on a fixed road the real car's vibration RMS rises with speed with elasticity 0.60 [0.50, 0.69] on urban and 0.85 [0.74, 0.93] on motorway segments, the simulated vibration over the same segments at the same speeds with 0.92–1.30 and 1.07–1.40 (Table S16). The simulator does not reproduce how the car's

TABLE III

SIMULATION AGAINST PAIRED REAL DRIVES ON THE SAME 20 DRAWS OF k KM OF TRAINING-ROUTE ROAD (POOLED SEGMENT MAE IN M/KM ON THE HELD-OUT ROUTES; BRACKETS: 2.5–97.5 % OF DRAWS). REAL-ONLY: LIGHTGBM OR THE THREE-PARAMETER VIBRATION INDEX TRAINED ON THE REAL PASSES OVER THE k KM. SAME k KM: LIGHTGBM TRAINED ON V0 SIMULATED PASSES OVER THE PROFILES OF THE SAME k KM, WITH THE PAIRED DIFFERENCE TO REAL-ONLY LIGHTGBM AND THE NUMBER OF THE 20 DRAWS IN WHICH SIMULATION WAS BETTER. ALL + LEVEL: LIGHTGBM TRAINED ON SIMULATED PASSES OVER ALL TRAINING-ROUTE PROFILES WITH A LEVEL OFFSET FROM THE k KM OF REAL PASSES, RANGE OF THE MEANS OVER THE NINE MEASURED-PROFILE VARIANTS V0–V8.

	Real-only		Same k km, simulated (V0)			All + level
k	LightGBM	index	MAE	− real	better	V0–V8
1	1.63 [1.08, 2.36]	1.30	1.33	−0.30	20	0.48–0.62
2	1.14 [0.63, 1.62]	0.94	0.86	−0.28	20	0.45–0.58
5	0.68 [0.50, 1.07]	0.72	0.60	−0.08	16	0.42–0.56
10	0.52 [0.45, 0.67]	0.71	0.49	−0.03	15	0.42–0.58
20	0.46 [0.43, 0.50]	0.69	0.47	+0.01	9	0.41–0.58
40	0.44 [0.42, 0.47]	0.68	0.45	+0.01	3	–
all	0.42	–	0.44	–	–	–

vibration scales with speed; as this mismatch is similar for all variants, it does not explain their different levels.

D. How much paired real data is needed

Fig. 4 and Table III compare simulation with paired real drives on the same 20 draws of k km of training-route road; Table S14 gives every variant. Two real-only comparators are used, LightGBM and the three-parameter vibration index, which suits very small samples better (1.30 vs 1.63 m/km at 1 km). The ranges over draws reflect which k km are drawn, not the uncertainty of the test routes.

Equal profile budget (Fig. 4a). Simulated drives over the profiles of the k km are not a general substitute for the real drives over them. With 1–2 km, simulated LightGBM beats real-trained LightGBM for every variant but not the real-trained index (simulation better in 3–10 of 20 draws at 1 km), and all these errors exceed 0.86 m/km. From 5 km on the answer depends on the simulator: at 20 km the difference to real-only LightGBM ranges from -0.02 to $+0.24$ m/km over the nine variants, small for V0 and V2 ($+0.01$ and -0.02) and largest for V1, V3 and V5, the variants with the largest 10–15 Hz deficit ($+0.22$–$+0.24$). A level offset from the same k km closes the gap of the single-vehicle variant V4 ($+0.01$ m/km at 20 km) but not that of these three.

Paired drives used only for the level (Fig. 4b). Trained on simulated drives over all profiled training roads and calibrated with a level offset from 5 km of paired drives, every variant reaches 0.42–0.56 m/km, better than real-only LightGBM on those 5 km (0.68 m/km) in 13–20 of 20 draws and better than the real-only index (0.72 m/km) in 19–20; real training on all data of the training routes reaches 0.42 m/km, the six variants other than V1, V3 and V5 0.42–0.45 m/km. The advantage at 5 km holds on each held-out route for every variant (mean difference per route -0.37 to -0.04 m/km). With more paired data, real-only LightGBM overtakes V1, V3 and V5: the calibrated simulation is better in only 3–5 of 20 draws at 10 km and in none at 20 km, while for the other six variants it is still better in 14–20 of 20 draws at 20 km. For these three variants a level offset is not enough; an affine calibration from 20 km brings them to 0.45–0.48 m/km. Simulation over all profiled roads plus a level from a few kilometres of paired drives was thus the most data-efficient option for every variant; how long it stays ahead of real drives depends on how closely the simulated vibration matches the car's.

E. Speed dependence on a fixed road

Each segment was driven about ten times at different speeds (median within-segment SD of log speed 0.12), so the speed dependence of every estimate can be measured on a road of constant roughness (Fig. 5, Table S17). The real car's vibration RMS rises by $+83$ % between 40 and 100 km/h (elasticity $+0.66$ [$+0.60$, $+0.71$]). The learned estimators and the vibration index trained on real cars over-correct for it: LightGBM's estimate falls by -13 % over the same range (-0.15 [-0.19, -0.11]), and without speed inputs it still falls (-0.14); the MLP and CNN behave alike (-0.19 and -0.19) and the vibration index over-corrects most (-0.64). Simulation-trained LightGBM V0 changes by -6 % (-0.06 [-0.11, -0.01]). Paired on the same passes, its elasticity exceeds that of the real-trained model by $+0.09$ [$+0.06$, $+0.12$] (5 km blocks), entirely on urban roads ($+0.12$ [$+0.09$, $+0.15$]; motorways $+0.02$ [-0.06, $+0.06$]). Across the nine variants the pooled elasticity ranges from -0.15 to $+0.06$ (8 of 9 inside the ± 10 % band), the paired difference from $+0.00$ to $+0.21$ (interval excluding zero for 8 of 9); on motorways every estimate falls with speed (simulation-trained -0.36 to -0.16, real-trained -0.29). Simulation thus removes part of the over-correction that real-trained estimators show on urban roads, but it does not make estimates speed-invariant, and the simulation-trained vibration index still over-corrects strongly (-0.48). Given the speed-scaling mismatch of Section V-C, this is a property of the learned estimators, not evidence that the simulator reproduces the car's speed response.

F. Other vehicles and sensors: simulation only

LiRA-CD contains one car model, so transfer to other vehicles can be examined only in simulation (class S, Table IV, Fig. 6). With the amount of training data fixed at 1,200 passes, the MAE on 30 unseen vehicles exceeds that on the training vehicles by $+70$ % with 3 training vehicles, $+16$ % with 30 (the number used in V0) and $+6$ % with 100. The criterion of our earlier study (upper bound ≤ 10 %) is met from 100 vehicles with a segment bootstrap but not with a route bootstrap (upper bounds 11 and 12 % for 100 and 300 vehicles). The gap closes from both sides: the error on unseen vehicles falls (1.06 to 0.87 m/km from 3 to 100 vehicles) and the error on training vehicles rises as each gets fewer passes (0.62 to 0.82 m/km). At $N = 30$, a vehicle's error is only weakly related to its parameter distance from the nearest training vehicle (Spearman 0.14 over 30 vehicles).

Sensor perturbations at test time (Table IV, bottom) reproduce in simulation what the variants showed on the real car:

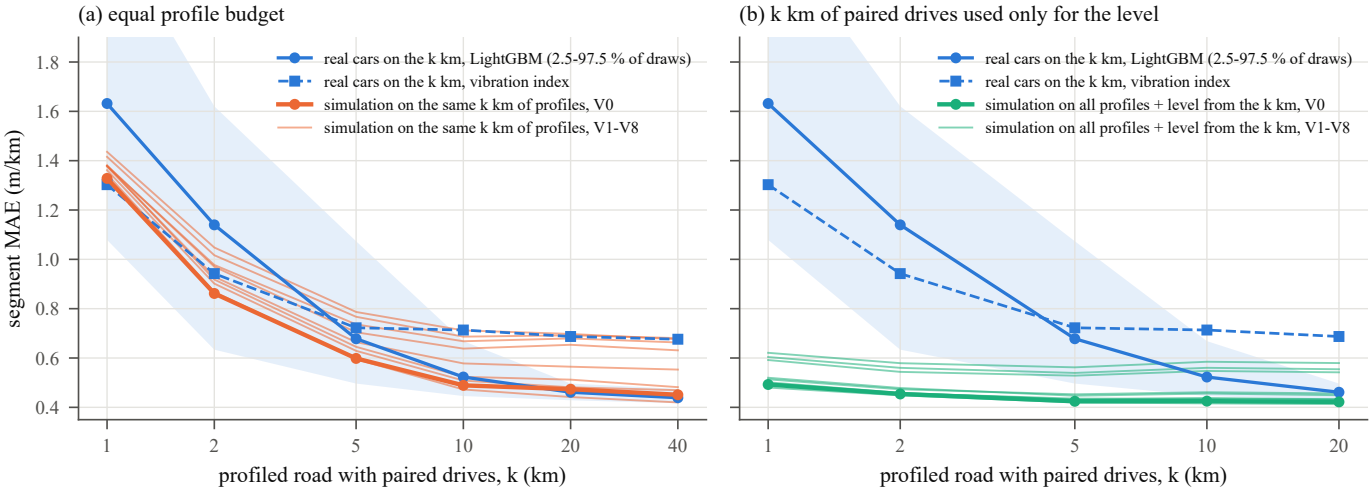


Fig. 4. Simulation against paired real drives on the same 20 draws of k km of training-route road (pooled segment MAE on the held-out routes). Blue: trained on the real passes over the k km, LightGBM (band: 2.5–97.5 % of draws, clipped at the top) or the three-parameter vibration index (dashed). (a) Equal profile budget: LightGBM trained on simulated passes over the profiles of the same k km (bold: V0; thin: V1–V8). (b) LightGBM trained on simulated passes over all training-route profiles, with a level offset estimated from the real passes over the k km.

TABLE IV

CONTROLLED SIMULATION (CLASS S; NO REAL DATA). LEFT: NUMBER OF TRAINING VEHICLES AT A FIXED 1,200 TRAINING PASSES (HELD-OUT SYNTHETIC ROUTES, 30 UNSEEN AND UP TO 30 SEEN VEHICLES; LIGHTGBM). Δ: UNSEEN MINUS SEEN MAE RELATIVE TO SEEN, WITH THE UPPER 95 % BOUND FROM A SEGMENT AND FROM A ROUTE BOOTSTRAP; CRITERION OF OUR EARLIER STUDY: UPPER BOUND $\leq$10 %. RIGHT: ONE SENSOR PROPERTY CHANGED AT TEST TIME (LIGHTGBM TRAINED ON SYNTHETIC PROFILES WITH THE SENSOR MIXTURE, TESTED ON 12 HELD-OUT SYNTHETIC ROUTES DRIVEN BY 6 UNSEEN VEHICLES WITH THE TELEMATICS-LIKE SENSOR; MAE AND BIAS IN M/KM).

Vehicles N	Seen MAE	Unseen MAE	Δ (%)	Upper bound (%) segments / routes	Criterion met segments / routes
3	0.62	1.06	+70	76 / 87	no / no
10	0.75	0.92	+24	27 / 31	no / no
30	0.78	0.90	+16	18 / 21	no / no
100	0.82	0.87	+6	8 / 11	yes / no
300	0.81	0.87	+7	9 / 12	yes / no

Sensor property at test time	MAE	Bias
telematics-like, rigid (reference)	0.84	−0.65
+ holder resonance 40 Hz	0.79	−0.55
+ holder resonance 25 Hz	0.73	−0.36
+ holder resonance 15 Hz	0.88	+0.37
sampling rate 200 Hz	0.77	−0.48
sampling rate 50 Hz	0.87	−0.71
sampling rate 25 Hz	1.39	−1.36
speed error +20 %	1.06	−0.98
speed error −20 %	0.73	−0.15
white noise ×8	0.82	−0.58
mounting tilt s.d. 20°	0.84	−0.64
40 % samples lost	0.94	−0.81

the sensor path moves the level. A holder resonance at 15 Hz turns a bias of −0.65 into +0.37 m/km; 25 Hz sampling and a speed error of +20 % are the most harmful perturbations (MAE 1.39 and 1.06 m/km), white noise, tilt and lost samples the least.

VI. DISCUSSION

What simulated drives over measured profiles provide. Ranking segments within a route turned out to be the easy part: an untrained, speed-normalised vibration RMS does it as well as any trained estimator, real or simulated. What training has to supply is the mapping from vibration to IRI in m/km across roads and speeds. For six of nine simulator variants, simulation supplied this mapping up to a level offset; the level itself depends on the absolute gain of the car's vibration path, which none of the variants was built to reproduce for this car. For a road agency this suggests a two-step recipe: simulate the car type over roads it has already profiled, then fix the level with a few kilometres of paired drives. Before trusting a zero-shot level, the real and simulated spectra of the same calibration roads can be compared; in our data the 10–15 Hz amplitude ratio ordered the level errors of nine variants, but this relation was found after the fact on non-independent variants and needs confirmation on other cars.

A correction to our earlier explanation. Our earlier study attributed the transfer of V0 to the breadth of its sensor model. The 2×2 design contradicts this: V0's level came from the holder resonance in half of its passes, which happened to supply part of the 10–15 Hz energy that the simulated car lacks. V0's zero-shot accuracy is therefore a fortunate match, not a property of randomising over sensors, and we no longer report it as evidence that a simulator can be used without real data.

Speed. In naturalistic driving smooth roads are driven fast,

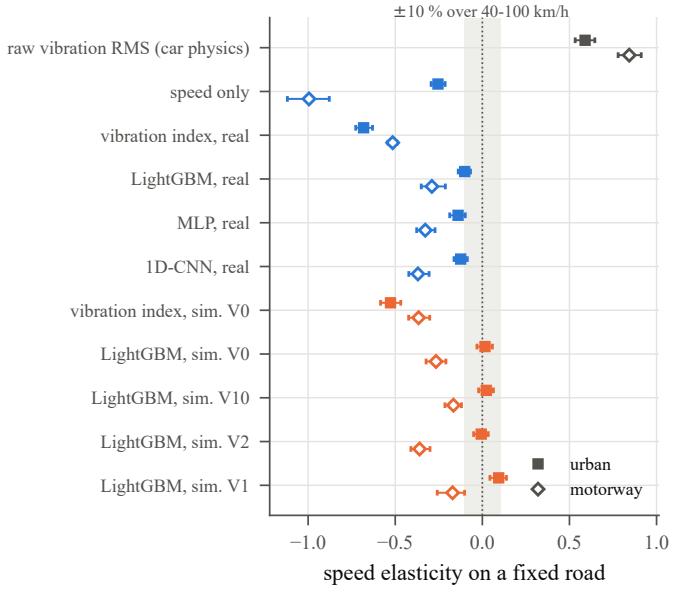


Fig. 5. Speed elasticity of each estimate on a fixed real road (slope of log estimate on log speed with segment fixed effects; 95 % intervals from a 1 km-block bootstrap; urban and motorway segments). Grey band: ±10 % change between 40 and 100 km/h. Top row: raw vibration RMS of the real car.

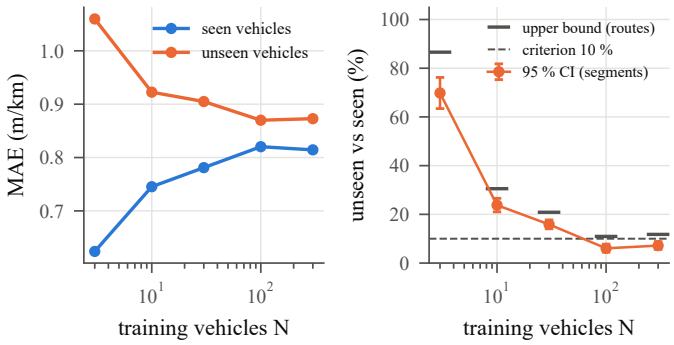


Fig. 6. Number of training vehicles at a fixed 1,200 training passes (simulation only). Left: MAE on seen and unseen vehicles. Right: relative increase from seen to unseen vehicles with 95 % interval (segment bootstrap) and the upper bound of a route bootstrap; dashed: 10 % criterion.

so estimators trained on real cars can use speed as a cue for low roughness, and on a fixed road their estimates fall with speed (Section V-E). Simulated speeds are drawn independently of roughness, and simulation-trained estimators over-correct less on urban roads; on motorways they over-correct as much as real-trained ones, and the simulated vibration itself scales with speed differently from the real car's. Repeated passes over the same segments make this measurable on real roads, and we recommend reporting such fixed-road elasticities instead of speed-sweep tests in simulation alone.

Statistics. Segment errors are spatially correlated, and 5 km-block intervals of paired differences were 1.9–2.7 times as wide as segment-level ones. A segment bootstrap would have declared V0 worse than the real-trained model; the block bootstrap does not, and the route-level interval with four routes is wider still ([−0.06, +0.10] m/km). Even 5 km blocks leave some correlation (0.13 at 5 km), mostly from route-level offsets, and route-level intervals include zero for most comparisons (Table S5); the supported differences therefore describe these four routes, not a population of routes. Benchmarks on road data should resample spatial blocks or routes, not segments.

Reference uncertainty. All errors are relative to one P79 survey. A second device (ARAN9000, surveyed on other dates) agrees with it in ranking (Spearman 0.84 urban, 0.95 motorway) and in median level (ratio 1.07 and 1.12), but disagrees strongly on a few urban segments (5 % above 15 m/km), which dominate its mean disagreement (Table S11). We therefore do not compare estimator errors with device disagreement.

VII. Limitations

External validity. LiRA-CD contains one car model with one type of telematics unit on four routes in one city region; road class and route are confounded, and with four routes no statement about route-to-route variation is inferential. Whether the simulated mapping transfers up to a level offset, and how much paired data the level needs, for other cars, sensors, speeds and road networks is unknown. Transfer to other vehicles and to phones in holders was examined only in simulation. *Evidence class.* Except for the V0 comparison, all real-data results of Sections V-B–V-E come from analyses specified after the test routes had been used, some (the sensor factorial, the spectral diagnosis, the speed scaling) after an independent review had pointed at the sensor model; we report every variant and draw, but these analyses generate hypotheses rather than confirm them. *Simulator.* The simulator is a linear half-car without roll or body flexibility, driven by the mean of the two wheel-path profiles, with parameter ranges for sensitivity rather than a calibrated fleet. Its vibration differs from the real car's at 2–5 Hz and 10–20 Hz and in its speed scaling, and we could not identify the physical source of the 10–15 Hz excess (sensor mounting, unsprung dynamics, tyre enveloping and roll are all candidates). *Baselines.* M7 follows the idea of calibrated quarter-car inversion [10] but is not a reproduction, because the vehicle parameters calibrated in that study are not available; the CNN is the architecture of our earlier study without tuning. *Other.* The reference surveys were not simultaneous with the car passes, segments are 100 m long, and we did not train on real data from several cities.

VIII. Conclusion

Simulated drives over measured road profiles taught gradient boosting to rank the road segments of a real car as well as real drives did, under every sensor and vehicle model we tried, but an untrained vibration statistic ranks them as well; the informative test is the level. The simulated level was not reliable: it depended on the simulated vibration at 10–15 Hz, and the simulated vibration grew faster with speed than the real car's. Combined with a level offset from 5 km of paired drives, simulation over all profiled roads was more accurate than training on those 5 km alone for every simulator variant; with 10 km or more of paired drives its advantage depended on

the variant. For a new car type this suggests simulating over roads that are already profiled, calibrating the level with a few kilometres of paired drives and checking the simulated against the real vibration spectrum before relying on the simulator. Whether these findings hold for other vehicles, sensors and road networks requires new paired data, ideally collected after freezing the simulator.

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